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Student Number

Year 12, 2023 HSC Mathematics Advanced Assessment Task 1 (38 Marks)

Sequences and Series

Series and Finance

Discrete Probability Distributions

Graphs and Equations

Staff Involved:

8:30 AM

DXC JAI LAK LMD ARM DZB*

PDJ ARP ALY RJW

WEDNESDAY 8th FEBRUARY

155 copies

General

Instructions:

- Working time - 55 minutes
- Write using black pen
- NESA approved calculators may be used
- A reference sheet is provided separately
- Show relevant mathematical reasoning and/or calculations
- Marks may not be awarded for careless or poorly arranged working

1. Circle all tables below that are **NOT** valid discrete probability distributions.

2 marks

x	0.1	0.2	0.3	0.4
$P(X = x)$	0.2	0.3	0.5	0.1

x	1	2	3	4
$P(X = x)$	0.25	0.25	0.25	0.25

x	5	10	15	20
$P(X = x)$	-0.1	0.2	0.4	0.5

x	-9	-3	3	9
$P(X = x)$	0.5	0	0	0.5

2. The overall financial return in dollars on a game of chance is shown in the complete discrete probability table below. Positive x values indicate financial gain, and negative x values indicate financial loss.

x	-4	0	3	9
$P(X = x)$	$4a$	$3a$	$2a$	a

i) Find a as a decimal.

1 mark

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ii) Calculate the expected value of the distribution.

Would you expect to make money playing this game?

2 marks

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- [illegible]

- [illegible]

- 1 mark**

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- 2 marks**

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- $$S_n = \frac{n}{2}[2a + (n - 1)d]$$

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- 2 marks**

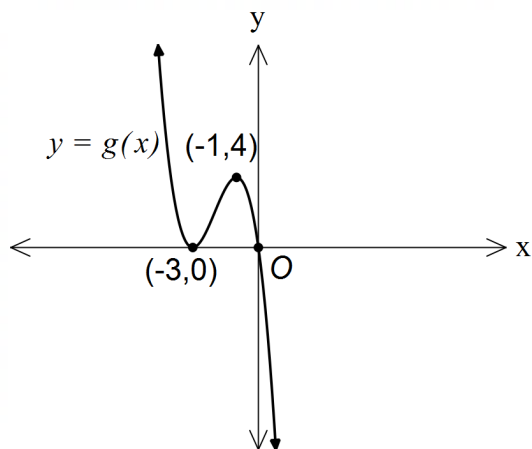
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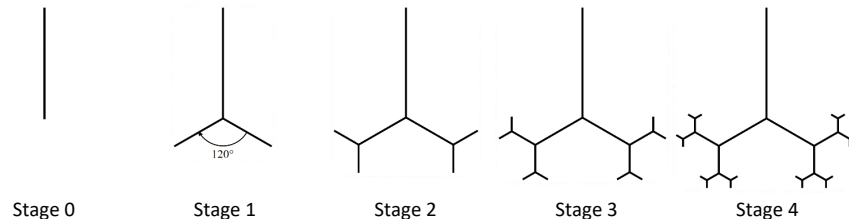
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7. Consider the function $f(x) = (x + 2)^2$. Sketch the function $y = f(x - 3) - 1$, labelling the vertex and all intercepts. **2 marks**

8. The graph of $y = g(x)$ is shown below. On the same axis, sketch the function $y = g(\frac{-x}{2})$, showing the coordinates of any x -intercepts and coordinates of any stationary points **2 marks**



9. Below is an example of a fractal, different to the one in your investigation.



The initial root (at Stage 0) is vertical and of length 1. The additional branches at each stage are half as long as the previous branch, and grow at 120° from each other, each a 60° rotation from the previous branch. The increase in vertical height of the total system in stage 1 is

$$\frac{1}{2} \cos 60^\circ = \frac{1}{4}. \text{ (You do NOT need to prove this)}$$

- i) Calculate the total vertical height of the system at Stage 3. **2 marks**

- ii) Find the limiting total vertical height of the fractal system. **3 marks**

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- 3 marks**

[illegible]

- 3 marks**

[illegible]

- i) Show that the amount owing after 2 months immediately after the instalment is paid is: **2 marks**

[illegible]

- 3 marks**

This image shows a full page of white paper with horizontal dashed lines, typical of primary school writing paper. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

Student number

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iii) The interest for each month, I , can be represented as

$$I = A_n \times r,$$

where A_n is the amount owing after the previous instalment and r is the monthly interest rate.

If Zoe made a monthly instalment where 50% of it erased the monthly interest and 50% of it went towards reducing the principal, calculate the amount still owing immediately after the instalment is paid.

2 marks

End of Paper

12 Adv 2023 Ass1 Solutions

①	x	0.1	0.2	0.3	0.4
	P(x)	0.2	0.3	0.5	0.1

has sum of $P(x) = 1.1$ ∴ not a D.P.O

x	5	10	15	20
P(x)	0.1	0.2	0.4	0.5

is not $(P(x) \geq 0)$.

② (i) $\sum P(x) = 1$

$$4a + 3a + 2a + a = 10a = 1$$

$$\therefore a = 0.1$$

$$i) E(X) = \sum xP(x)$$

$$= (-4)^0 \times 0.4 + 3^0 \times 0.2 + 9^0 \times 0.1$$

$$= -0.1$$

∴ No you will not make money

$$iii) Var(X) = \sum E(X^2) - \mu^2$$

$$= (-4)^2 \times 0.4 + 3^2 \times 0.2 + 9^2 \times 0.1 - (-0.1)^2$$

$$= 16.29$$

$$\therefore \sigma = \sqrt{16.29} = 4.036 \text{ (3dp)}$$

(iv)	x	-8	0	6	18
	P(x)	0.4	0.3	0.2	0.1

$$xP(x) = -3.2 + 0 + 1.2 + 1.8 = -0.2$$

∴ worse scenario

$$③ a = 4, d = 5, T_{10} = a + 9d$$

$$= 4 + 45 = 49$$

$$④ ar = 12, ar^4 = 15$$

$$\frac{②}{①} \text{ gives } = \frac{15}{12} = r^3$$

$$\therefore r^3 = \frac{5}{4}, r = \frac{1}{2}$$

$$⑤ L = T_n = a + (n-1)d$$

$$\therefore S_n = \frac{n}{2}(a + L)$$

$$= \frac{n}{2}(a + a + (n-1)d)$$

$$= \frac{n}{2}(2a + (n-1)d)$$

$$⑥ r = \frac{T_2}{T_1} = \frac{2k}{3} \text{ Limiting sum exists for}$$

$$|r| < 1 \text{ i.e. } \left| \frac{2k}{3} \right| < 1$$

$$\therefore -1 < \frac{2k}{3} < 1$$

$$-3 < 2k < 3$$

$$-\frac{3}{2} < k < \frac{3}{2}$$

$$⑦ f(x) = (x+2)^2$$

$$f(x-3) - 1 = ((x-3)+2)^2 - 1$$

$$= (x-1)^2 - 1$$

$$\therefore \text{vertex} = (1, -1)$$

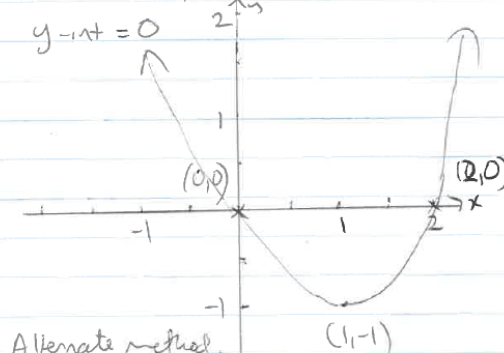
$$x\text{-ints when } y = 0$$

$$\therefore (x-1)^2 - 1 = 0$$

$$(x-1)^2 = 1$$

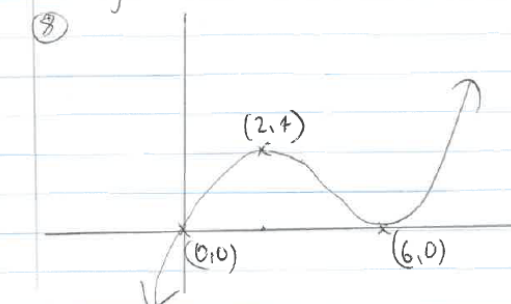
$$x-1 = \pm 1$$

$$x = 1 \pm 1 \text{ i.e. } x = 0 \text{ or } 2.$$



Alternate method.

Sketch $y = (x+2)^2$ then translate 3 right and one down.



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9(c) Vertical height

$$S_1 \quad 1 + \frac{1}{4}$$

$$S_2 \quad 1 + \frac{1}{4} + \frac{1}{4} = 1\frac{1}{2}$$

$$S_3 \quad y = \frac{1}{8} \cos 60^\circ = \frac{1}{16}$$



$$\therefore \text{vertical height} = 1\frac{1}{2} + \frac{1}{16} = 1\frac{9}{16}$$

(ii) Limiting height

$$= 1 + \frac{1}{4} + \frac{1}{4} + \frac{1}{16} + \frac{1}{16} + \frac{1}{32} + \frac{1}{32} + \dots$$

$$= 1 + 2\left(\frac{1}{4} + \frac{1}{16} + \frac{1}{32} + \dots\right)$$

$$S_n, a = \frac{1}{4}, r = \frac{1}{4}$$

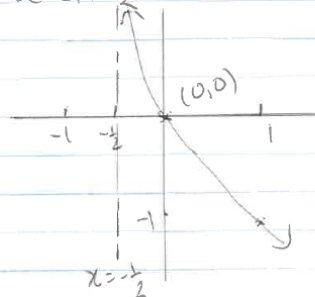
$$\therefore S_n = \frac{\frac{1}{4}}{1 - \frac{1}{4}} = \frac{1}{3}$$

$$\therefore \text{limiting height} = 1 + 2\left(\frac{1}{3}\right) = \frac{5}{3}$$

10 Reflect in x-axis

Scale horiz $\times \frac{1}{2}$

Move left $\frac{1}{2}$



$$\textcircled{1} T_5 = a + 4d = 4 \quad \textcircled{1}$$

$$T_2 + T_5 + T_{10} = a + d + 4 + a + 9d = 16$$

$$2a + 10d = 12$$

$$a + 5d = 6 \quad \textcircled{2}$$

$$a + 4d = 4 \quad \textcircled{1}$$

$$\textcircled{2} - \textcircled{1}$$

$$d = 2$$

Continued. Sub $d = 2$ into $\textcircled{1}$

$$T_5 = a + 4d = 4$$

$$a + 8 = 4$$

$$a = -4$$

$$\therefore T_{10} = a + 9d$$

$$= -4 + 18 = 14$$

$$\textcircled{12} A_1 = 760000(1.0025) - 3604.01 \quad (r = \frac{0.025}{12} = 0.0025)$$

$$A_2 = A_1 \times 1.0025 - 3604.01$$

$$= (760000(1.0025) - 3604.01) \times 1.0025 - 3604.01$$

$$= 760000(1.0025)^2 - 3604.01(1.0025) - 3604.01$$

$$= 760000(1.0025)^n - 3604.01(1 + 1.0025)$$

$$\textcircled{ii} A_n = 760000(1.0025)^n$$

$$- 3604.01(1 + 1.0025 + \dots + 1.0025^{n-1})$$

$$n = n, r = 1.0025, a = 1 \text{ in } S_n$$

and $A_n = 0$ when loan is paid off.

$$\therefore 0 = 760000(1.0025)^n - 3604.01 \left(\frac{1.0025^n - 1}{0.0025} \right)$$

$$\therefore 1441604 \times 1.0025^n - 1441604$$

$$= 760000(1.0025)^n$$

$$\therefore 681604 \times 1.0025^n = 1441604$$

$$1.0025^n = \frac{1441604}{681604} = 2.115$$

$$\therefore \ln 1.0025 = n = 299.99 \dots$$

$$\ln \left(\frac{1441604}{681604} \right) = 299.99 \dots \therefore 300 \text{ months} = 25 \text{ years}$$

$$\textcircled{iii} I = A_n \times r, r = 0.0025, I = \frac{1}{2}(3604.01)$$

$$\therefore 1802.005 = A_n \times 0.0025$$

$$\therefore A_n = \$720802 \text{ is amount owing}$$

after previous instalment.

$\therefore$ Amount owing after this instalment is

$$A + I - M = 720802 + 1802.005 - 3604.01$$

$$= 718999.99$$